INCH-POUND
MIL-E-24715
10 March 1989

MILITARY SPECIFICATION

EXTINGUISHERS, FIRE, VAPORIZING LIQUID (BROMOCHLORODIFLUOROMETHANE, HALON 1211)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rechargeable fire extinguishers of the stored pressure type charged with bromochlorodifluoromethane (Halon 1211).

1.2 Classification. Halon 1211 fire extinguishers shall be of the following sizes and types, as specified (see 6.2):

Type I - Hand portable extinguisher, magnetic (for size 20)
Type II - Hand portable extinguisher, non-magnetic (for size 20)
Type III - Wheeled extinguisher (for size 150)
Type IV - Extinguisher with bracket (for size 150)
Size 20 - 20-pound capacity Halon 1211 extinguisher
Size 150 - 150-pound capacity Halon 1211 extinguisher

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-S-300 - Sheeting and Tape, Reflective: Nonexposed Lens.
BB-N-411 - Nitrogen, Technical.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4210
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MIL-P-116	- Preservation, Methods of.
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-I-17214	- Indicator, Permeability; Lo-Mu (Go-No Go).
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.
MIL-P-27401	- Propellant Pressurizing Agent, Nitrogen.
MIL-B-38741	- Bromochlorodifluoromethane, Technical.
MIL-E-24269	- Extinguisher, Fire, Carbon Dioxide, 15 Pound, Portable, Permanent Shutoff, Navy Shipboard Use.
MS51336	- Lunette - Coupler, Drawbar, Ring.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-H28/2	- Screw-Thread Standard for Federal Services, Section 2 Unified Inch Screw Threads - UN and UNR Thread Forms.
FED-STD-313	- Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.
FED-STD-595	- Colors.

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-2073-1	- DoD Materiel Procedures for Development and Application of Packaging Requirements.
DOD-STD-2142	- Magnetic Silencing Characteristics, Measurement of (Metric).
DOD-STD-2143	- Magnetic Silencing Requirements for the Construction of Nonmagnetic Ships and Craft (Metric).

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government drawing and publication. The following other Government drawing and publication form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
810-1385843 - Holder Device, CO_2 Fire Equipment.

(Application for copies should be addressed to the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

PUBLICATION

DEPARTMENT OF TRANSPORTATION (DOT)
Code of Federal Regulations (CFR), Title 49.
Part 173 - Shippers General Requirements for Shipment and Packaging.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 484 - Standard Specification for General Requirements for Stainless and Heat-Resisting Bars, Billets and Forgings. (DoD adopted)
B 117 - Standard Method of Salt Spray (Fog) Testing. (DoD adopted)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

COMPRESSED GAS ASSOCIATION (CGA)
C-14 - Procedures for Fire Testing of DOT Cylinder Safety Relief Device Systems.

(Application for copies should be addressed to the Compressed Gas Association, Inc., 1235 Jefferson Davis Highway, Arlington, VA 22202.)

UNDERWRITERS LABORATORIES, INC. (UL)
ANSI/UL 711 - Fire Extinguishers, Rating and Fire Testing of.
ANSI/UL 1093 - Halogenated Agent Fire Extinguishers.

(Application for copies should be addressed to Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Standard product. Accessories and components furnished under this specification shall be in the same component assembly and of the same quality as furnished commercially with the standard product, except for deviations required to meet this specification (see 6.3).

3.3 Extinguisher. The extinguisher shall be a stored-pressure type. The extinguisher shall be listed and labeled by UL in accordance with ANSI/UL 711 and ANSI/UL 1093 for Halon 1211. The extinguishers for type I shall also be listed as "marine type". The size 150 extinguisher, which shall be UL listed and labeled when charged, does not need to be UL labeled when shipped empty.

3.3.1 Fire extinguishing agent. The extinguishing agent shall be bromochlorodifluoromethane (Halon 1211) as specified in MIL-B-38741. The extinguisher shall be super-pressurized with dry nitrogen in accordance with BB-N-411, grade A or B, or MIL-P-27401, grade A, B, or C.

3.3.2 Materials. Working parts (such as valves, springs and pins) in the discharge or actuating assemblies shall be made of nonferrous metal, or austenitic stainless steel in accordance with ASTM A 484 which shall withstand frequent exposure to Halon 1211 without damage. Aluminum shall not be used in any component with the exception of the nozzle and type II extinguishers.

3.3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials as opposed to virgin raw material. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3.3 Finish.

3.3.3.1 Painting. The cylinder, mounting bracket, and chassis shall be coated with an enamel or powder paint protective coating in accordance with best commercial practice for enhanced corrosion resistance.

3.3.3.2 Color code. The extinguisher shall be color coded red gloss number 11105 in accordance with FED-STD-595. A reflective silver band, type II, class 1, with a reflectivity of 2 in accordance with L-S-300, shall be attached all around the cylinder. The band shall be clearly marked in black lettering "Halon 1211" on the front of extinguisher. The minimum band width shall be 2 inches and the minimum letter size 1 inch.

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3.3.4 Threads. Threads shall be in accordance with FED-STD-H28 and FED-STD-H28/2.

3.3.5 Fasteners. Metal fasteners, such as nuts, bolts, washers, studs, and rivets, and clip exposed to the atmosphere shall be austenitic stainless steel.

3.4 Extinguisher components.

3.4 1 Components for type I and II extinguishers. The type I and II extinguisher shall have the following major components:

(a) Discharge valve assembly.
(b) Discharge hose and nozzle assembly.
(c) Cylinder.
(d) Mounting bracket.
(e) Pressure relief device.

3.4.1.1 Discharge valve assembly. The discharge valve assembly shall be a self-closing, handle and lever, squeeze type. The handle and lever mechanism shall stop the flow of agent when the lever is released. The discharge valve assembly shall consist of valve body, siphon tube, operating lever and carrying handle, pressure gauge, and pull pin and retainer.

3.4.1.1.1 Valve body and siphon tube. The valve body and siphon tube shall be metallic.

3.4.1.1.2 Pressure gauge. A pressure gauge as specified in ANSI/UL 1093 shall be provided as an integral part of the discharge valve assembly. The gauge shall indicate the pressure in units of pounds per square inch (lb/in^2). The gauge case shall be vented and closed with a tape or grommet and shall be moisture tight.

3.4.1.1.3 Operating lever and carrying handle. The metallic lever and handle shall be attached to the valve body by pins or rivets. Pins or rivets shall be spun over or otherwise closed on both ends.

3.4.1.1.4 Pull pin and retainer chain. The discharge valve shall have a safety pin that prevents inadvertent operation. The pin shall pass through the operating lever in such a way that the operating lever will not operate with the pin in place. The pin shall be held captive to the extinguisher by a short stainless steel or non-ferrous metallic retainer chain or lanyard, so that it will not be lost during operation of the unit. A visual mechanical tamper seal shall be used to hold the pull pin in the safe position. Use of, or tampering with the extinguisher shall break the seal. The seal shall be field-replaceable.

3.4.1.2 Discharge hose and nozzle assembly. The discharge hose and nozzle assembly shall be provided in accordance with ANSI/UL 1093. The length of the discharge hoses, including the nozzle, shall be not less than 16 inches and not more than 22 inches. The nozzle shall be provided on the end of the discharge hose. The nozzle shall be metallic or fabricated of tough, rigid temperature and water resistant material. A metallic clip shall retain the nozzle or discharge hose next to the cylinder when not in use.

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3.4.1.3 Cylinder. The cylinder shall be of metallic construction in accordance with CFR Title 49, 173.306(c) or 173.37 (DOT 3AA). The dimensions of the cylinder shall be as follows:

Diameter: 6-3/4 ± 1/4 inches.
Service pressure: 195 lb/in^2.
Minimum wall thickness: 0.075 inch.
Maximum height: 27 inches.
Material: Steel (for type I); aluminum or stainless steel (for type II).

3.4.1.4 Mounting bracket. The extinguishers shall be provided with a heavy duty vehicle type mounting bracket. The bracket shall have a bottom plate which extends beyond the base of the fire extinguisher. The bracket, when closed, shall securely hold the cylinder and discharge hose, and nozzle assembly. The bracket shall be adjustable to retain cylinders of the diameter range specified.

3.4.1.4.1 Shipboard backfit mounting bracket. When specified (see 6.2), the mounting bracket shall be configured to be mechanically attached without welding to existing shipboard 15-pound CO_2 fire extinguisher mounting bracket fittings as specified on Drawing 810-1385843 or Aeroequip MBS 62028 that meets the requirements of MIL-E-24269. The bracket shall be adjustable to retain cylinders of the diameter range specified. The bracket shall be a heavy duty vehicle type.

3.4.1.5 Pressure relief device. The extinguisher shall be fitted with a rupture disk type pressure relief device that shall rupture at not less than 495 lb/in^2 and not more than 585 lb/in^2. The device shall consist of a rupture disk, rupture disk washer, and rupture disk nut or rupture nut assembly.

3.4.2 Components for type III and IV extinguishers. The type III and IV extinguishers shall have the following components:

(a) Discharge valve assembly
(b) Discharge hose
(c) Nozzle
(d) Cylinder
(e) Hose retainer
(f) Chassis (for type III only)
(g) Bracket (for type IV only)

3.4.2.1 Discharge valve assembly. The discharge valve assembly shall consist of a valve, pressure gauge, siphon tube, and pull pin and retainer chain.

3.4.2.1.1 Valve. The valve shall be a metallic lever which operates from fully open to fully closed with 90 degrees of lever rotation.

3.4.2.1.2 Siphon tube. The siphon tube shall be installed such that the agent shall be discharged completely with the cylinder in both the upright and horizontal positions.

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3.4.2.1.3 Pressure gauge. A pressure gauge as specified in ANSI/UL 1093 shall be provided as an integral part of the discharge valve assembly. The gauge shall indicate the pressure in units of lb/in^2. The pressure range indicated on the gauge shall be from 0 to 400 lb/in^2. The gauge case shall be vented and closed with a tape or an elastomeric grommet and shall be moisture tight.

3.4.2.1.4 Pull pin and retainer chain. The pull pin and retainer chain shall be provided as required for type I and II extinguishers (see 3.4.1.1.4).

3.4.2.2 Discharge hose. The discharge hose shall be provided in accordance with ANSI/UL 1093. The hose length shall be 50 feet and 75 feet for type III and IV, respectively.

3.4.2.3 Nozzle. A nozzle shall be installed at the end of the discharge hose. The nozzle shall have a shutoff mechanism to permit intermittent discharge and shall be configured to optimize discharge range. The nozzle shall be made of metallic material and shall be liquid-tight in the closed position. The nozzle shall incorporate a swivel at the hose connection to prevent kinking of the hose during deployment.

3.4.2.4 Cylinder. The cylinder shall be DOT 4BW or DOT 3AA in accordance with CFR Title 49. The dimensions of the cylinder shall be as follows:

Diameter: 14 ± 1/4 inches
Service pressure: 240 lb/in^2
Minimum volume: 4200 cubic inches
Maximum height: 39 inches

3.4.2.4.1 Pressure relief device. The extinguisher shall be fitted with a rupture disk type pressure relief device that shall rupture at not less than 400 lb/in^2 and not more than 480 lb/in^2. The device shall consist of a rupture disk, rupture disk washer, and rupture disk nut or a rupture disk nut assembly.

3.4.2.4.2 Agent level indicator. A gauge for determining the amount of agent in the extinguisher shall be provided. The level indicator shall operate with plus or minus 10 percent of correct level when the ambient temperature is minus 40 to plus 120 degrees Fahrenheit (°F). Accurate level indicator shall operate only when the unit is in the normal vertically upright position.

3.4.2.5 Hose retainer. The hose retainer shall retain the discharge hose and nozzle. The hose retainer shall allow coiling of the hose around an upper and lower hose support and come off easily, without kinking, when pulled from the upright stowed position.

3.4.2.6 Chassis (for type III only). The chassis frame shall be formed of round steel tubing with continuously welded joints to protect the tube interior from the environment. The frame shall carry all structural load.

3.4.2.6.1 Coupler and wheels (for type III only). The frame for type III extinguishers shall include a handle located so that one person can easily push and pull the extinguisher. The wheels shall be mounted on steel axles located at the center of balance of a fully charged extinguisher. The wheels shall be

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equipped with semi-pneumatic rubber tires. The coupler (as specified in MS51336) with wheels shall be provided. The units shall be equipped with a lubrication fitting. The wheels with rubber track shall be at least 16 inches in diameter.

3.4.2.7 Bracket (for type IV only). The bracket for type IV extinguisher may be an integral part of the cylinder or may be a separate saddle assembly. A saddle assembly shall consist of two saddle clamps and bolts and nuts required for securing the saddle clamps to the saddle. The saddle assembly shall have a clip to support the cylinder when the clamps are loosened. The bracket shall have four bolt holes to permit attachment to a flat surface.

3.5 Weight and UL ratings. The extinguisher shall successfully extinguish test fires in accordance with ANSI/UL 711 and shall meet the following requirements:

Size	Maximum full weight (pounds)	Minimum UL classification ratings
20 (Type I, II)	38	4A:80B:C
150 (Type III)	340	30A:240B:C
150 (Type IV)	280	

3.6 Performance.

3.6.1 Fire extinguishment. The extinguisher shall have a minimum fire extinguishing rating as specified in 3.5.

3.6.2 Discharge range. The extinguisher shall have an initial minimum discharge range in still air as follows (see 4.6.2):

Size	Range (feet)
20	15
150	25

3.6.3 High-impact shock resistance. The extinguisher shall discharge not less than 90 percent of rated agent capacity and shall show no sign of damage and leakage after being subjected to high-impact shock (see 4.6.3). Retention of the discharge hose and nozzle by the nozzle clip or hose retainer is not required. The extinguisher shall be securely retained by the mounting bracket.

3.6.4 Vibration resistance (for type I and II). The extinguisher shall discharge not less than 90 percent of rated agent capacity and shall show no sign of damage and leakage after being subjected to vibration (see 4.6.4). The extinguisher shall be securely retained by the mounting bracket.

3.6.5 Corrosion resistance. The extinguisher and bracket shall show no signs of corrosion after exposure to salt spray (see 4.6.5). The extinguisher shall discharge not less than 90 percent of rated agent capacity.

3.6.6 Nonshatterability. The cylinder shall not break into more than two pieces when struck by a 0.50 caliber projectile (see 4.6.6).

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3.6.7 Magnetic permeability (for type II only). The type II extinguisher and mounting bracket relative magnetic permeability after fabrication of all material shall not exceed 2.0 (see 4.6.7).

3.6.8 Operation of mounting bracket (for type I and II). The mounting bracket of type I and II extinguisher shall be operable and retain the extinguisher after being subjected to the test specified in 4.6.8.

3.6.9 Pressure relief device performance. The extinguishers shall pass the acceptance criteria of CGA C-14 (see 4.6.9).

3.7 Shipping. When specified (see 6.2), the extinguisher shall be shipped:

(a) Fully charged and pressurized for normal operation, or
(b) Without extinguishing agent, size 150 only.

3.8 Markings. The extinguisher shall be marked as specified in ANSI/UL 1093 for Halon 1211 fire extinguishers. The labels shall be metal plates with pressure-sensitive adhesive backing, securely attached to the extinguisher. The plates shall not become detached and the printing shall be legible after the extinguisher has been subjected to all the tests specified herein.

3.8.1 Military identification. A label plate shall be affixed to the cylinder in accordance with MIL-STD-130 and shall include the following:

(a) Manufacturer's name.
(b) Commercial and Government Entity (CAGE).
(c) Manufacturer's part number.
(d) National stock number, when specified (see 6.2).
(e) Contract number.
(f) Military specification number, type, size.

A separate label shall be used to show any of the above information which is not contained on the primary extinguisher nameplate.

3.8.2 Weight. The gross charged weight of the extinguisher and the weight of the agent shall be shown on the label plate.

3.8.3 Hydro test date and manufacturer serial number. The date at which the cylinder was hydrostatically tested and manufacturer serial number shall be stamped into the outer surface of a permanent part of the cylinder in 1/4 inch minimum height letters.

3.8.4 Pressure table (for size 150 only). A label shall be affixed to the cylinder showing normal charged Halon 1211 pressures at temperatures from minus 40 to plus 120°F.

3.8.5 Wall thickness. The wall thickness of the cylinder shall be stamped "wall thick _____ inch" on the top portion of cylinder (curvature area) with thickness to the nearest one thousandth inch. (For example, "wall thick .075 inch") the stamp letters shall be not less than 1/4 inch high.

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3.9 Recharging. Extinguishers shall be rechargeable and recharging shall be accomplished without replacement of any valve part.

3.10 Material safety data sheet (MSDS). The contracting activity shall be provided a material safety data sheet at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.7).

3.11 Workmanship. The extinguisher shall be constructed, assembled, and furnished in a manner to assure good quality hardware and an overall neat appearance. Particular attention shall be given to evidence of proficiency and quality control such as: freedom from blemishes, burrs, defects, and sharp edges; accuracy of dimensions, radii of fillets, and markings of the assembly; thoroughness of welding, brazing, riveting, and painting; and alignment of parts and tightness of assembly screws and bolts. The extinguisher shall be thoroughly cleaned of all foreign material, including moisture.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. Except where otherwise specified, two extinguishers of each type and size shall be examined and tested in accordance with the first article inspections specified in table I. If one of the sample extinguishers fails any test specified in table I, the extinguisher represented for first article inspection shall be rejected (see 6.3).

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TABLE I. Inspections.

Inspection	Requirement	Test	First article inspection	Quality conformance inspection
Visual examination		4.5	X	X
Discharge range	3.6.2	4.6.2	X	
Shock	3.6.3	4.6.3	X	
Vibration	3.6.4	4.6.4	X	
Salt spray	3.6.5	4.6.5	X	
Gunfire	3.6.6	4.6.6	X	
Magnetic permeability (type II only)	3.6.7	4.6.7	X	X
Mounting bracket	3.6.8	4.6.8	X	
Pressure relief device	3.6.9	4.6.9	X	

4.4 Quality conformance inspection. The sample extinguishers selected in accordance with 4.4.2 shall be subjected to the quality conformance inspections specified in table I. Any sample which fails any quality conformance inspection shall be rejected (see 6.3).

4.4.1 Inspection lot. For the purpose of quality conformance inspection, those extinguishers of the same class and section shall be considered a lot.

4.4.2 Sampling for quality conformance. Five sample extinguishers shall be selected at random from each lot.

4.5 Visual examination. Each sample extinguisher selected in accordance with 4.4.2 shall be visually and dimensionally examined to determine conformance to this specification. The sample shall be examined for defects as specified in table II. Any sample failing to conform to the visual and dimensional examination shall be rejected.

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TABLE II. Defects.

Categories	Defects
101	Missing UL label or lack of other evidence of conforming to UL requirements.
102	Materials and components not as specified.
103	Finish of exterior surfaces not as specified.
104	Weights not as specified.
105	Dimensions not as specified.
106	Marking, operational, or maintenance instruction not as specified.
107	Damaged components or evidence that the extinguisher is inoperable.
108	Workmanship not as specified.

4.6 Tests. The tests as specified in 4.6.1 through 4.6.9, except 4.6.7 shall be conducted with the extinguisher charged to the rated capacity with Halon 1211.

4.6.1 General. Except for size 150 extinguishers shipped empty (see 3.3), all extinguishers shall be labeled with the UL label. (The UL label or approval means that the extinguisher has passed all testing specified in ANSI/UL 1093 for Halon 1211 extinguishers.)

4.6.2 Discharge range. Each extinguisher shall be discharged in still air. A measurement shall be made during the first 10 seconds of discharge at 70 ± 5°F ambient temperature to determine the distance at which the visible liquid phase of the discharge stream changes into vapor phase (see 3.6.2).

4.6.3 Shock. Each charged type I and II extinguisher shall be weighed and locked into the mounting bracket. Each charged type III and IV extinguisher shall be weighed and attached to the test apparatus. The assembly shall then be subjected to shock testing for grade A, class I, type A, lightweight equipment, as specified in MIL-S-901. At the conclusion of the shock test, the extinguisher shall stand for a minimum of 10 minutes and then be weighed to determine if any leakage has occurred. It shall then be discharged as evidence of proper operation. The discharge shall be interrupted at least once by use of the discharge valve (see 3.6.3).

4.6.4 Vibration. Each charged type I and II extinguisher shall be locked in the mounting bracket and then subjected to the type I vibration test specified in MIL-STD-167-1 for shipboard equipment (see 3.6.4).

4.6.5 Salt spray. Each extinguisher shall be locked in its mounting bracket or chassis and placed upright in a salt spray solution in accordance with ASTM B 117. Duration of the salt spray test shall be 480 hours (see 3.6.5).

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4.6.6 Gunfire. Each cylinder shall be charged to the rated service pressure plus or minus 5 percent using a nonliquified gas. The cylinders shall be placed behind a steel barricade. The cylinder shall be in such a position that a bullet passing through a hole in the barricade strikes the cylinder at right angles to the longitudinal centerline within 1 inch of the longitudinal centerline and near to the vertical center of the cylinder. The cylinder temperature at the time of the test shall be between 50 and 100°F. An armor-piercing projectile 0.50 caliber in size shall be fired at the cylinder. The 0.50 caliber projectile shall strike the cylinder at a velocity of 2800 ± 100 feet per second. The projectile shall strike the cylinder straight on (not tumbled). A cylinder shall be considered as having failed this test if the cylinder breaks into more than two pieces, provided, however, that pieces smaller than 2 inches in diameter coming from the areas (centering on the perforation and 4 inches in diameter) on the cylinder adjacent to the point of entry and exit of the projectile are not counted. Cylinder, representative samples of which have passed this test, shall be marked "Nonshat" to indicate this fact (see 3.6.6).

4.6.7 Magnetic permeability (for type II only). The magnetic permeability of each item or assembly shall be non-magnetic in accordance with DOD-STD-2143, determined by means of a permeability indicator of the go-no go type conforming to or equal to that specified in MIL-I-17214 and shall be tested in accordance with test method P01 of DOD-STD-2142. The magnetic permeability shall conform to 3.6.7.

4.6.8 Mounting bracket (for type I and II). The extinguisher shall be locked in the mounting bracket and the bracket shall be opened. This procedure shall be repeated for 100 cycles within 30 minutes (see 3.6.8).

4.6.9 Pressure relief device. Three extinguishers shall be tested in accordance with CGA C-14 (see 3.6.9).

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

5.1.1.1 Lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

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Level A and B - Type II - weather resistant.
Category 1 - general use.
Level C - Type I - non-weather resistant.
Category 1 - general use.

5.1.1.2 Fiberboard. Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes shall meet the flammability and smoke requirements of PPP-F-320.

5.2 Preservation. Preservation shall be level A, C, or commercial as specified (see 6.2).

5.2.1 Level A. Fire extinguishers shall be individually unit protected in accordance with method III of MIL-P-116. The unit containers shall conform to PPP-B-636, class weather-resistant, type SF with style optional to the contractor. Containers shall be closed in accordance with method V of the appendix to PPP-B-636.

5.2.2 Level C. Fire extinguishers shall be individually unit protected as specified for level A except that unit containers may be of the domestic class. Closure shall conform to method I using pressure sensitive tape in accordance with the appendix to PPP-B-636.

5.2.3 Commercial. Preservation shall be in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C or commercial, as specified (see 6.2).

5.3.1 Levels A, B, and C containers. Fire extinguishers preserved as specified (see 5.2) shall be packed in shipping containers, for the level specified (see 6.2), in accordance with the appendix of MIL-STD-2073-1 and herein. Unless otherwise specified (see 6.2), container selection shall be at the contractor's option.

5.3.1.1 Caseliners, closure, and gross weight.

5.3.1.1.1 Caseliners. Unless otherwise specified (see 6.2), level A shipping containers with fire extinguishers preserved level C or commercial shall be provided with waterproof caseliners in accordance with MIL-STD-2073-1.

5.3.1.1.2 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method V of PPP-B-636 and reinforced with non-metallic or tape banding and domestic fiberboard boxes shall be closed in accordance with method I of PPP-B-636 using pressure sensitive tape.

5.3.1.1.3 Weight. Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

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5.3.2 Commercial. Fire extinguishers preserved as specified (see 5.2) shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.2.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two, 3- by 4-inch nominal wood skids laid flat, or a skid or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, storage, and stowage.

5.4 Palletized unit loads. Shipping containers shall be palletized in accordance with MIL-STD-2073-1, when specified (see 6.2).

5.5 Marking. In addition to any special marking required (see 6.2 and herein), interior (unit) packs, exterior shipping containers and palletized loads shall be marked for shipment, stowage and storage in accordance with MIL-STD-2073-1 and the appendix thereto. Marking shall include bar coding.

5.5.1 Caution tag. Unless otherwise specified (see 6.2), the extinguishers covered by this specification shall be shipped in a charged condition. If the fire extinguishers are specified to be uncharged when shipped, a caution tag shall be attached to each fire extinguisher prior to packaging in the unit container with the following information:

"CAUTION

This fire extinguisher is not charged. Follow instructions on the extinguisher for charging."

5.5.2 Additional marking (for type II only). Unless otherwise specified (see 6.2), the following special marking on the shipping container is required for type II extinguishers "Non-magnetic material - protect from magnetic contamination".

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The extinguishers covered by this specification are intended for use on board ships where a clean, non-corrosive agent is needed to prevent contamination and residue. The extinguishers are intended for use on class A, B, and C fires.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) If first article inspection is required (see 3.1).

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(e) When shipboard backfit mounting bracket is required (see 3.4.1.4.1).
(f) Extinguisher shipped as specified (see 3.7).
(g) When National Stock Number is required (see 3.8.1(d)).
(h) When fire-retardant material is not required (see 5.1.1.1).
(i) Level of preservation (see 5.2).
(j) Level of packing (see 5.3 and 5.3.1).
(k) When container selection is not at the contractor's option (see 5.3.1).
(l) When waterproof caseliners are not required (see 5.3.1.1.1).
(m) When palletization is required (see 5.4).
(n) Special marking required (see 5.5 and 5.5.2).
(o) When extinguishers are not shipped in a charged condition (see 5.5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID number	DID title	Suggested tailoring
3.2	DI-E-7031	Drawings, engineering and associated lists	Level 3
4.3	DI-T-4902	First article inspection report	-----
4.3	DI-T-4901	First article inspection procedure	-----
4.4	DI-T-5329	Inspection and test reports	-----

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

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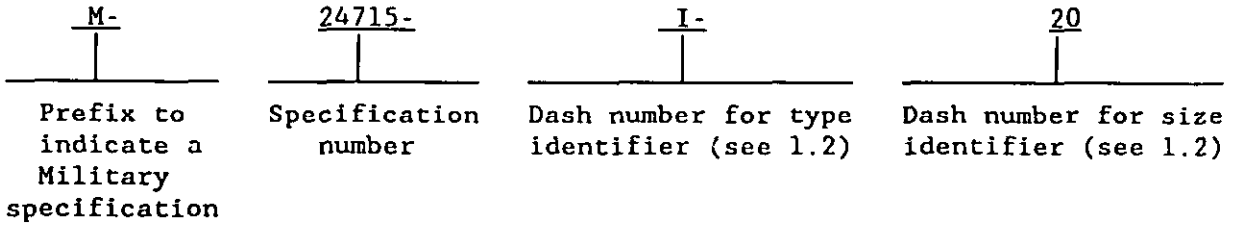
6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a production sample, a first article sample, a first production item, a sample selected from the first _____ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Material safety data sheets (MSDS). Contracting officers must identify those activities requiring copies of MSDS's. Additional required Government information is contained in FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract.

6.8 Part number. The Part or Identifying Number (PIN) consists of the letter "M" followed by the specification number and assigned dash numbers as follows (see 1.2):

Example:



6.9 Subject term (key word) listing.

Cylinder
Discharge hose
Nozzle
Pressure-relief device

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-24715	EXTINGUISHERS, FIRE, VAPORIZING LIQUID (BROMOCHLORODIFLUOROEMETHANE, HALON 1211)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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